

# SOC Analyst Training Lab

## Endpoint Event Monitoring & Log Analysis

Windows 11 Endpoint · Wazuh SIEM · Event ID Analysis

Prepared by: KHIZAR UL ISLAM

Cybersecurity / SOC Home Lab Exercise

### Lab Environment

Windows 11 endpoint — Khizar-Malik-Windows11 (agent ID 001, IP 192.168.44.1) · Monitoring via Wazuh Threat Hunting / Discover · Alerts index wazuh-alerts-4.x · Windows Security auditing (logon + process creation) active on the endpoint.

## 1. Lab Objective

---

Perform common Windows endpoint activities — login (success/fail), Command Prompt commands, and a GUI app — then check Wazuh to see which ones generated a matching alert. This mirrors the core SOC L1 task of correlating endpoint behavior with SIEM detections.

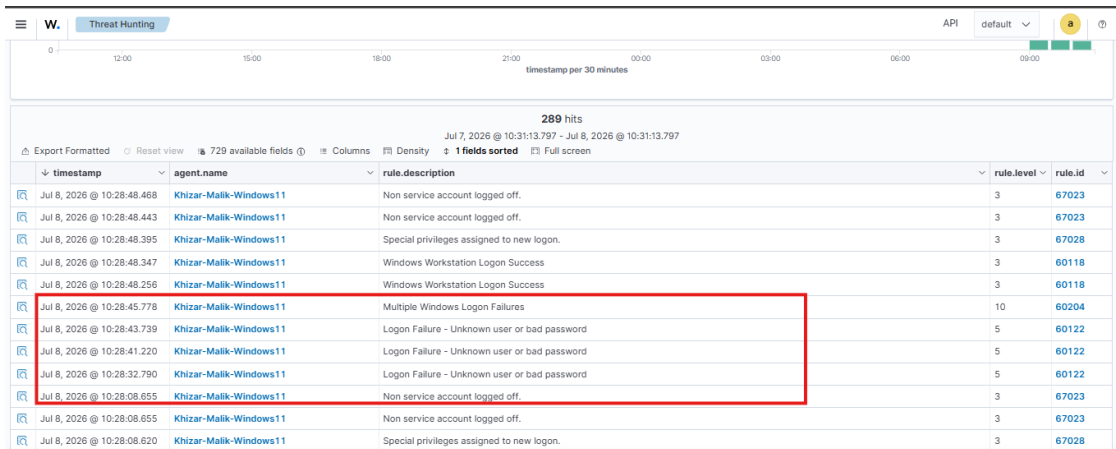
### Key Principle

Not all activities trigger alerts. Wazuh only fires an alert when an incoming event matches a defined rule. Understanding this gap is fundamental to SOC work.

## 2. Activities Performed & Alerts Observed

### 2.1 Login Events

Wazuh was checked for login-related events on the endpoint, covering both failed and successful logon attempts.



| timestamp                  | agent.name             | rule.description                             | rule.level | rule.id |
|----------------------------|------------------------|----------------------------------------------|------------|---------|
| Jul 8, 2026 @ 10:28:48.468 | Khizar-Malik-Windows11 | Non service account logged off.              | 3          | 67023   |
| Jul 8, 2026 @ 10:28:48.443 | Khizar-Malik-Windows11 | Non service account logged off.              | 3          | 67023   |
| Jul 8, 2026 @ 10:28:48.395 | Khizar-Malik-Windows11 | Special privileges assigned to new logon.    | 3          | 67028   |
| Jul 8, 2026 @ 10:28:48.347 | Khizar-Malik-Windows11 | Windows Workstation Logon Success            | 3          | 60118   |
| Jul 8, 2026 @ 10:28:48.256 | Khizar-Malik-Windows11 | Windows Workstation Logon Success            | 3          | 60118   |
| Jul 8, 2026 @ 10:28:45.778 | Khizar-Malik-Windows11 | Multiple Windows Logon Failures              | 10         | 60204   |
| Jul 8, 2026 @ 10:28:43.739 | Khizar-Malik-Windows11 | Logon Failure - Unknown user or bad password | 5          | 60122   |
| Jul 8, 2026 @ 10:28:41.220 | Khizar-Malik-Windows11 | Logon Failure - Unknown user or bad password | 5          | 60122   |
| Jul 8, 2026 @ 10:28:32.790 | Khizar-Malik-Windows11 | Logon Failure - Unknown user or bad password | 5          | 60122   |
| Jul 8, 2026 @ 10:28:08.655 | Khizar-Malik-Windows11 | Non service account logged off.              | 3          | 67023   |
| Jul 8, 2026 @ 10:28:08.655 | Khizar-Malik-Windows11 | Non service account logged off.              | 3          | 67023   |
| Jul 8, 2026 @ 10:28:08.620 | Khizar-Malik-Windows11 | Special privileges assigned to new logon.    | 3          | 67028   |

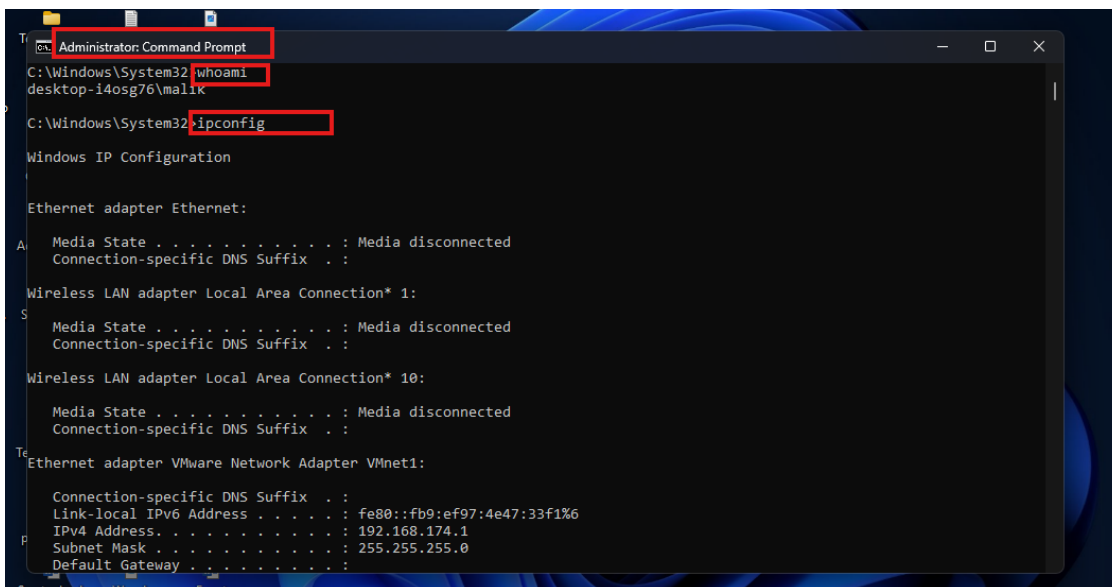
Figure 1 — Wazuh Discover view showing login events: failed logons, multiple failures, successful logons, and logoff, all for agent Khizar-Malik-Windows11.

#### Alert Highlight — Rule 60204 (Level 10)

Multiple Windows Logon Failures — Wazuh aggregates repeated failed logon events (rule 60122) and escalates to a level 10 alert. This pattern mirrors a brute-force attempt and would trigger investigation in a real SOC environment.

### 2.2 Command Prompt — whoami & ipconfig

whoami and ipconfig were run from Command Prompt to check user context and network configuration.



```
Administrator: Command Prompt
C:\Windows\System32>whoami
desktop-i4osg76\malik

C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 10:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

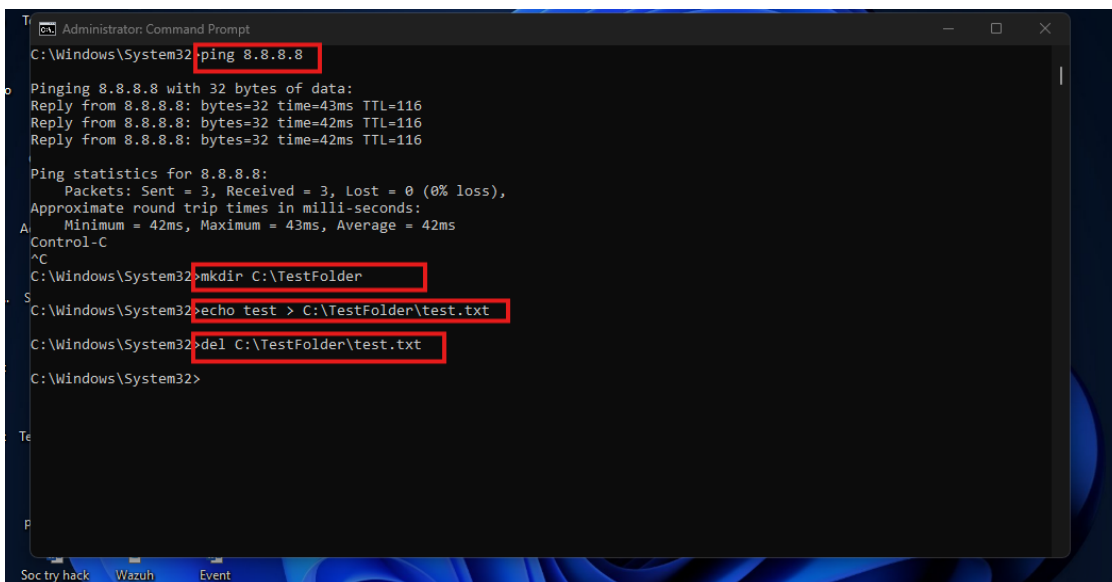
Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::fb9:ef97:4e47:33f1%6
IPv4 Address. . . . . : 192.168.174.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

Figure 2 — Command Prompt output showing whoami (desktop-i4osg76\malik) followed by ipconfig.

## 2.3 Command Prompt (Administrator) — ping, mkdir, file create/delete

A second, elevated Command Prompt was used to run ping, create a folder, create a file, and delete it.



```
Administrator: Command Prompt
C:\Windows\System32>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=43ms TTL=116
Reply from 8.8.8.8: bytes=32 time=42ms TTL=116
Reply from 8.8.8.8: bytes=32 time=42ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 42ms, Maximum = 43ms, Average = 42ms

C:\Windows\System32>mkdir C:\TestFolder

C:\Windows\System32>echo test > C:\TestFolder\test.txt

C:\Windows\System32>del C:\TestFolder\test.txt

C:\Windows\System32>
```

Figure 3 — CMD showing ping 8.8.8.8, mkdir C:\TestFolder, echo test > test.txt, and del test.txt.

## 2.4 Wazuh Alerts — Process Creation Overview

Every command above launches a new process, so Wazuh logged each one under a single rule.

| timestamp                  | agent.name             | rule.description       | rule.level | rule.id |
|----------------------------|------------------------|------------------------|------------|---------|
| Jul 8, 2026 @ 10:58:22.402 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:22.395 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:22.380 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:18.374 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.426 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.386 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.386 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.377 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.376 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:16.371 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:12.464 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:12.459 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:11.396 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:10.421 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |
| Jul 8, 2026 @ 10:58:07.419 | Khizar-Malik-Windows11 | A process was created. | 3          | 67027   |

Figure 4 — Wazuh Events list: repeated "A process was created" alerts (rule 67027) right after the commands were run.

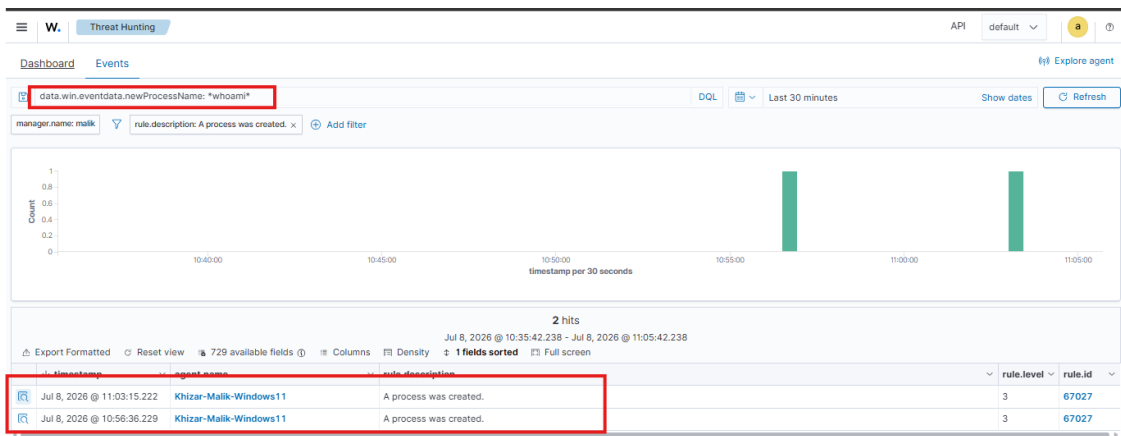


Figure 5 — Wazuh Threat Hunting filtered to rule.description "A process was created" for this agent, last 30 minutes.

## 2.5 Confirming whoami.exe

The raw alert document was opened to confirm the exact process captured.

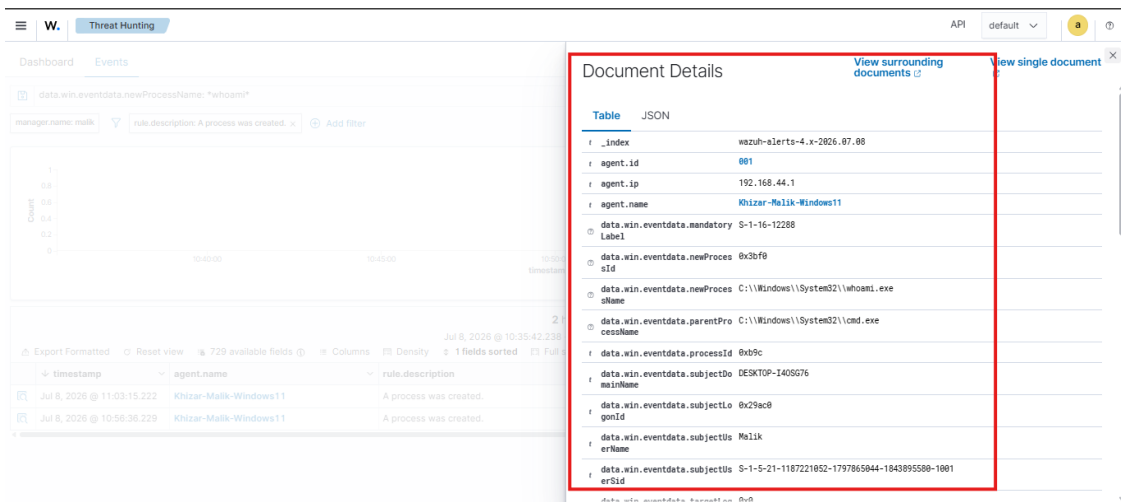


Figure 6 — Document details: New Process Name = whoami.exe, Parent Process Name = cmd.exe.

## 2.6 More Process Creation Alerts

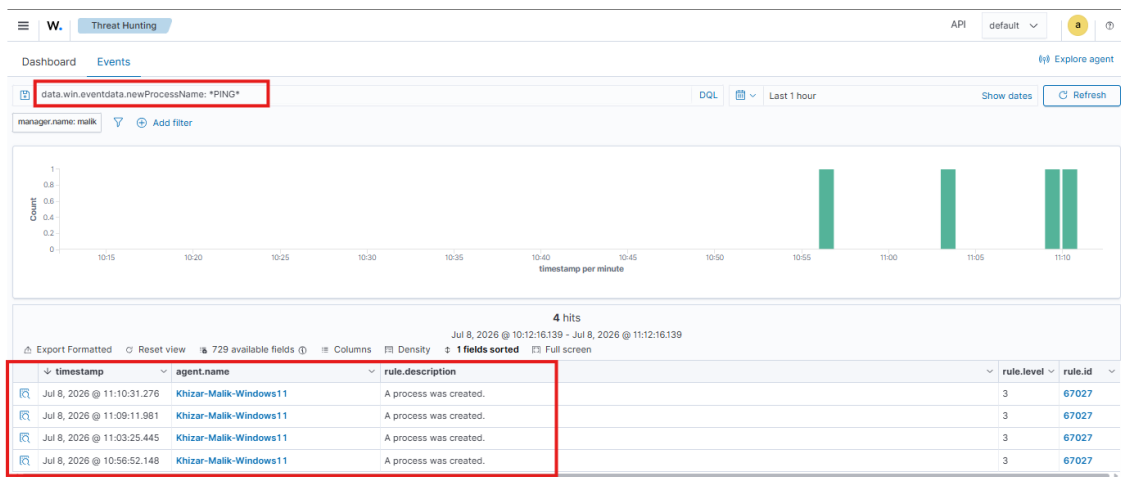


Figure 7 — Wazuh Events, last 1 hour: more "A process was created" alerts (rule 67027) for this agent.

2.7 Confirming ping.exe

Raw Windows event fields behind one alert were checked directly.

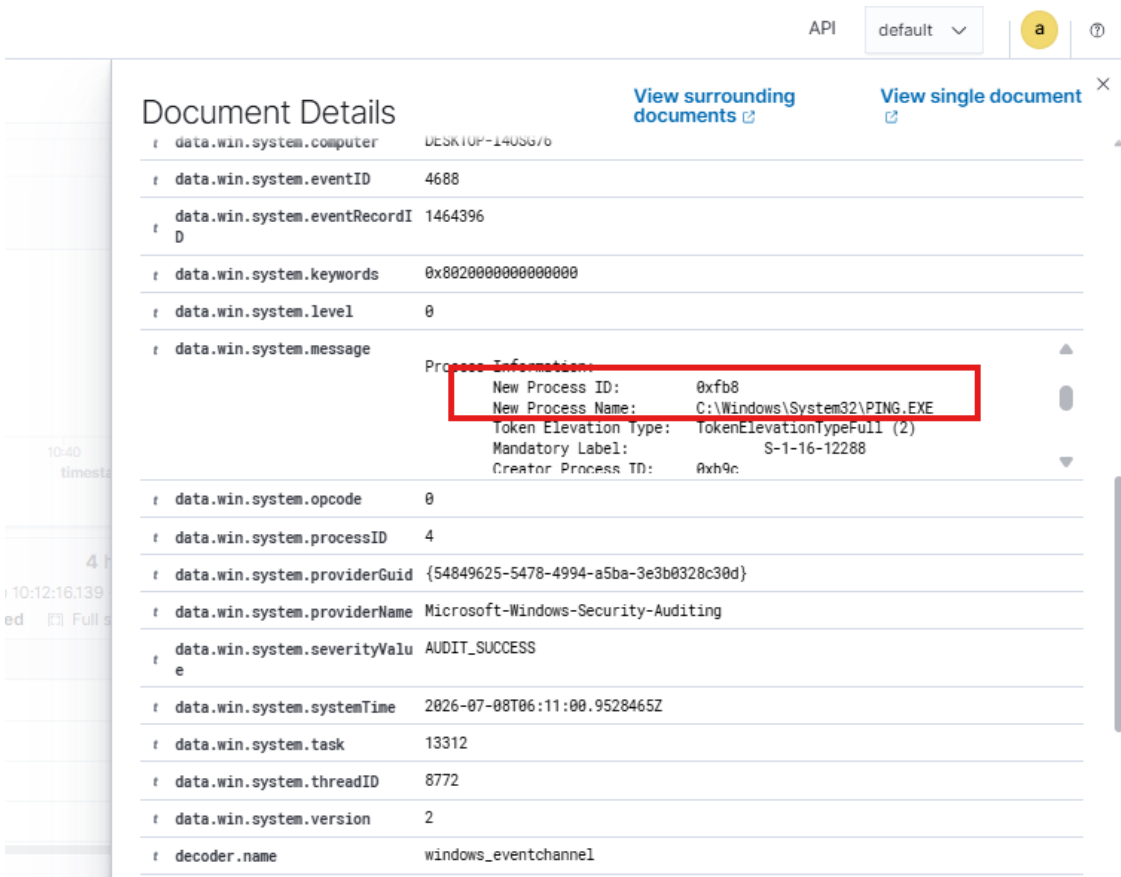


Figure 8 — Raw event details: New Process Name = PING.EXE, Event ID 4688, Provider = Microsoft-Windows-Security-Auditing.

2.8 Confirming ipconfig.exe

The ipconfig process was isolated with a targeted filter, then confirmed in the raw event.

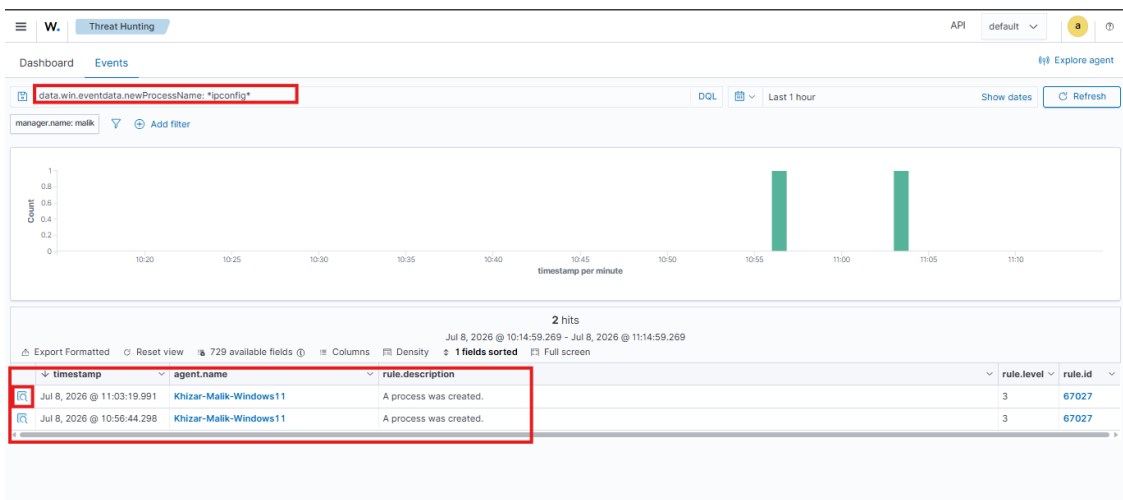


Figure 9 — Wazuh Events filtered on newProcessName: "ipconfig" — 2 hits found.

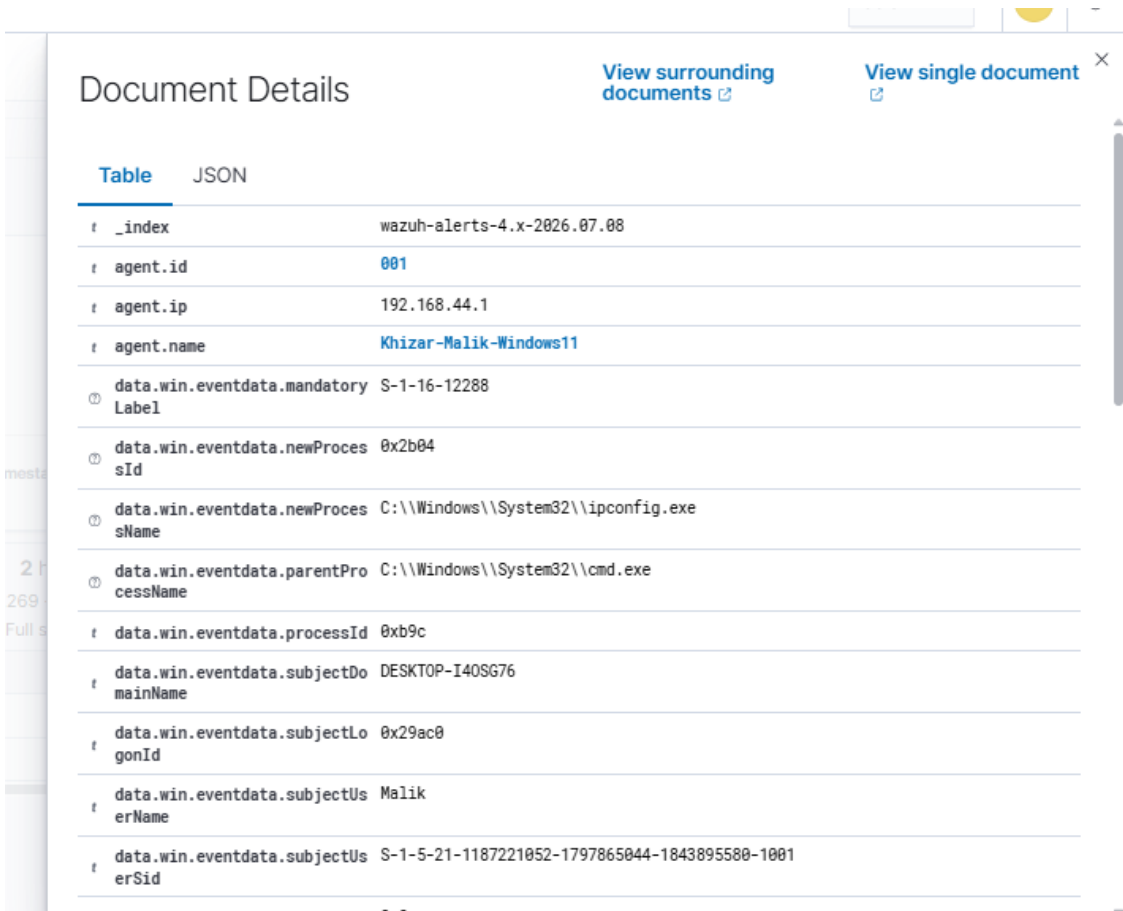


Figure 10 — Document details: New Process Name = ipconfig.exe, Parent Process Name = cmd.exe.

Document Details [View surrounding documents](#) [View single document](#)

|                               |                                                                                                                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data.win.system.eventRecordID | 1464018                                                                                                                                                                                                                                                                     |
| data.win.system.keywords      | 0x8020000000000000                                                                                                                                                                                                                                                          |
| data.win.system.level         | 0                                                                                                                                                                                                                                                                           |
| data.win.system.message       | Process Information:<br>New Process ID: 0x2b04<br>New Process Name: C:\Windows\System32\ipconfig.exe<br>Token Elevation type: TokenElevationTypeFull (2)<br>Mandatory Label: S-1-16-12288<br>Creator Process ID: 0xb9c<br>Creator Process Name: C:\Windows\System32\cmd.exe |
| data.win.system.opcode        | 0                                                                                                                                                                                                                                                                           |
| data.win.system.processID     | 4                                                                                                                                                                                                                                                                           |
| data.win.system.providerGuid  | {54849625-5478-4994-a5ba-3e3b0328c30d}                                                                                                                                                                                                                                      |
| data.win.system.providerName  | Microsoft-Windows-Security-Auditing                                                                                                                                                                                                                                         |
| data.win.system.severityValue | AUDIT_SUCCESS                                                                                                                                                                                                                                                               |
| data.win.system.systemTime    | 2026-07-08T06:03:48.8334223Z                                                                                                                                                                                                                                                |
| data.win.system.task          | 13312                                                                                                                                                                                                                                                                       |
| data.win.system.threadID      | 9432                                                                                                                                                                                                                                                                        |
| data.win.system.version       | 2                                                                                                                                                                                                                                                                           |
| decoder.name                  | windows_eventchannel                                                                                                                                                                                                                                                        |
| id                            | 1783490599.3291012                                                                                                                                                                                                                                                          |
| input.type                    | log                                                                                                                                                                                                                                                                         |

Figure 11 — Raw event fields for ipconfig.exe: Creator Process = cmd.exe, channel Security.

## 2.9 More Process Creation Alerts (Last 1 Hour)

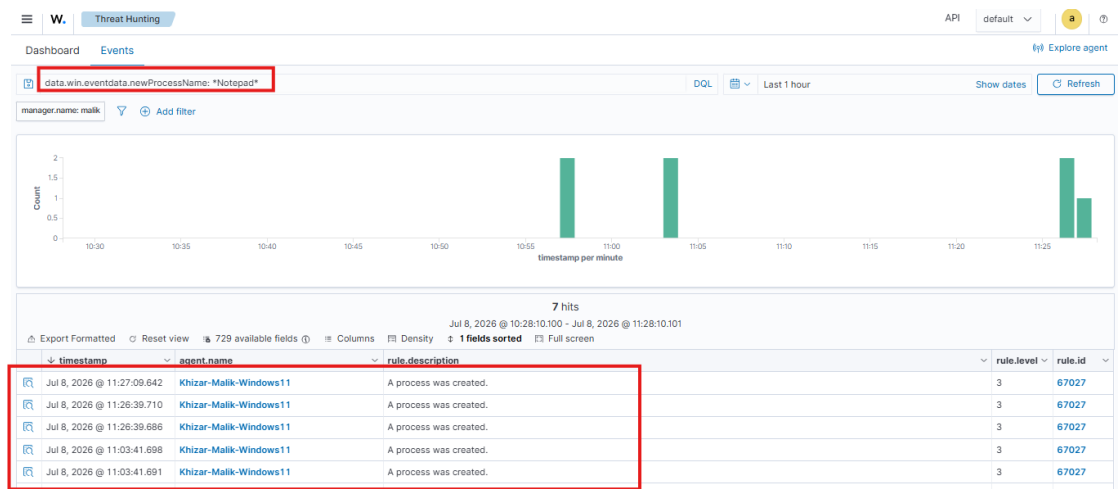


Figure 12 — Wazuh Events, last 1 hour: multiple "A process was created" alerts, one per command run.

## 2.10 Confirming Notepad.exe

Notepad was opened, then checked in Wazuh for the matching process creation event.



API

default

Document Details

View surrounding documents

View single document

Table

JSON

|                                      |                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| index                                | wazuh-alerts-4.x-2026.07.08                                                                                    |
| agent.id                             | 001                                                                                                            |
| agent.ip                             | 192.168.44.1                                                                                                   |
| agent.name                           | Khizar-Malik-Windows11                                                                                         |
| data.win.eventdata.mandatoryLabel    | S-1-16-12288                                                                                                   |
| data.win.eventdata.newProcessId      | 0x3bd8                                                                                                         |
| data.win.eventdata.newProcessName    | C:\\Program Files\\WindowsApps\\Microsoft.WindowsNotepad_11.2605.29.0_x64__8wekyb3d8bbwe\\Notepad\\Notepad.exe |
| data.win.eventdata.parentProcessName | C:\\Program Files\\WindowsApps\\Microsoft.WindowsNotepad_11.2605.29.0_x64__8wekyb3d8bbwe\\Notepad\\Notepad.exe |
| data.win.eventdata.processId         | 0x1988                                                                                                         |
| data.win.eventdata.subjectDomainName | DESKTOP-I40SG76                                                                                                |
| data.win.eventdata.subjectLogonId    | 0x29ac0                                                                                                        |
| data.win.eventdata.subjectUserName   | Malik                                                                                                          |
| data.win.eventdata.subjectUserSid    | S-1-5-21-1187221052-1797865044-1843895580-1001                                                                 |
| data.win.eventdata.timestamp         | 0x0                                                                                                            |

Figure 13 — Document details: New Process Name = Notepad.exe (Windows Store / WindowsApps version).

| Document Details                      |                                                                                                                                                                                                                                                                                                  | View surrounding documents | View single document |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------|
| data.win.eventdata.targetLogonId      | 0x0                                                                                                                                                                                                                                                                                              |                            |                      |
| data.win.eventdata.targetUserSid      | S-1-0-0                                                                                                                                                                                                                                                                                          |                            |                      |
| data.win.eventdata.tokenElevationType | %1937                                                                                                                                                                                                                                                                                            |                            |                      |
| data.win.system.channel               | Security                                                                                                                                                                                                                                                                                         |                            |                      |
| data.win.system.computer              | DESKTOP-I40SG76                                                                                                                                                                                                                                                                                  |                            |                      |
| data.win.system.eventID               | 4688                                                                                                                                                                                                                                                                                             |                            |                      |
| data.win.system.eventRecordID         | 1463628                                                                                                                                                                                                                                                                                          |                            |                      |
| data.win.system.keywords              | 0x8020000000000000                                                                                                                                                                                                                                                                               |                            |                      |
| data.win.system.level                 | 0                                                                                                                                                                                                                                                                                                |                            |                      |
| data.win.system.message               | Process Information:<br>New Process ID: 0x3bd8<br>New Process Name: C:\Program Files\WindowsApps\Microsoft.WindowsNotepad_11.2605.29.0_x64__8wekyb3d8bbwe\Notepad\Notepad.exe<br>Token Elevation Type: TokenElevationTypeFull (2)<br>Mandatory Label: S-1-16-12288<br>Creator Process ID: 0x1988 |                            |                      |
| data.win.system.opcode                | 0                                                                                                                                                                                                                                                                                                |                            |                      |
| data.win.system.processID             | 4                                                                                                                                                                                                                                                                                                |                            |                      |
| data.win.system.providerGuid          | {54849625-5478-4994-a5ba-3e3b0328c30d}                                                                                                                                                                                                                                                           |                            |                      |
| data.win.system.providerName          | Microsoft-Windows-Security-Auditing                                                                                                                                                                                                                                                              |                            |                      |

Figure 14 — Raw event fields for Notepad.exe: Event ID 4688, channel Security, Provider = Microsoft-Windows-Security-Auditing.

### Learning Point

Every process launched — even everyday commands like whoami, ping, ipconfig, and GUI apps like Notepad — is logged by Windows as Event ID 4688 and matched by the same Wazuh rule 67027 ("A process was created"). Login events use a separate set of rules (60122, 60204, 67023) tied to Event IDs 4624/4625/4634.

### 3. Activity Summary Table

Complete mapping of all activities performed, their Windows event, and the Wazuh alert generated:

| Activity              | Command                             | Event ID | Rule / Level            | Alert Description                       |
|-----------------------|-------------------------------------|----------|-------------------------|-----------------------------------------|
| Failed login          | Lock screen (Win+L), wrong password | 4625     | 60122 / L5, 60204 / L10 | Logon Failure + Multiple Logon Failures |
| Successful login      | Lock screen, correct password       | 4624     | 60118 / L3              | Windows Workstation Logon Success       |
| Logoff                | Session logoff                      | 4634     | 67023 / L3              | Non service account logged off          |
| whoami                | cmd.exe                             | 4688     | 67027 / L3              | A process was created — whoami.exe      |
| ipconfig              | cmd.exe                             | 4688     | 67027 / L3              | A process was created — ipconfig.exe    |
| ping 8.8.8.8          | cmd.exe                             | 4688     | 67027 / L3              | A process was created — PING.EXE        |
| mkdir C:\TestFolder   | cmd.exe                             | 4688     | 67027 / L3              | A process was created — cmd.exe child   |
| echo > test.txt / del | cmd.exe                             | 4688     | 67027 / L3              | A process was created — cmd.exe child   |
| notepad               | Start Menu / Run                    | 4688     | 67027 / L3              | A process was created — Notepad.exe     |

#### Conclusion

All activities performed in this lab generated a matching Wazuh alert. Login events map to rules 60122 / 60204 / 67023, while every command-line and GUI process launch maps to rule 67027 (Event ID 4688). This confirms Wazuh's default Windows ruleset already covers the most common SOC L1 detection scenarios without extra configuration.

## 4. Key Learnings

---

### 4.1 SOC Workflow Demonstrated

- **Performed activities on a live Windows endpoint and correlated them to SIEM alerts — the core L1 analyst skill.**
- Used Wazuh Threat Hunting to search by process name, event ID, and rule description.
- Read raw event data from the Document Details view — process name, parent process, user context, and timestamps.
- Observed rule severity levels — Level 3 (informational), Level 5 (warning), Level 10 (high — multiple failures).

### 4.2 Technical Observations

- Event ID 4688 fires for every process created, including command-line tools and GUI apps — filtering by process name is essential in a real investigation.
- Wazuh's aggregation rules (e.g. rule 60204) escalate severity when multiple low-level events repeat — useful for spotting patterns like brute force.
- Not all user actions trigger alerts by default — only those matching Wazuh's active ruleset. Coverage gaps can be closed with Sysmon or custom rules.
- The Document Details view is the fastest way to confirm an alert matches the exact action taken, by checking process name and parent process.